

Course Syllabus - ChE 360 Process Control

Spring 2025 (Unique # 16115)

Course Description: CHE 360 is an integrative Chemical Engineering course, bringing together knowledge from the other disciplines in the curriculum, including Heat and Mass Transfer, Separations, Reaction Engineering, Process Design and Numerical Methods. Starting from the material studied in these courses, CHE 360 aims to provide students with knowledge and understanding of process dynamics, process models, process control, and control system analysis and design.

Prerequisites: CHE 322, CHE 253M, and CHE 354.

Lectures: MWF 09:00 AM – 10:00 AM **CPE 2.216**

Recitations: M 01:00 PM – 03:00 PM **UTC 4.104**

Instructor: Dr. Thomas A. Badgwell / tbadgwell@che.utexas.edu
Office Hours: M: 5:00 – 6:00 PM <https://utexas.zoom.us/my/tbadgwell>

Teaching Assistant: Ying Zheng / yingzheng@utexas.edu
Office Hours: TBD

Tutor: Roger El-Bitar / roger.el-bitar@utexas.edu
Office Hours: TBD

Grader: Ruth Yao / ruth.y@utmail.utexas.edu

Course website: on Canvas <http://canvas.utexas.edu>

Req. Text: *Process Dynamics and Control*, Thomas F. Edgar, Duncan A. Mellichamp, Francis Doyle, John Wiley & Sons Inc, 4th Edition, (2016), available through the Longhorn Textbook Access (LTA) program.

Course Score (100 points total):

Exam Average (Three Midterms)	60%
Quiz Average (~10/11 Quizzes)	10%
Homework Average (~11 HWs)	30%

Course Grade (Course Score will be rounded to the nearest integer to determine grade):

90% <= A- < 94% <=	A	<= 100%
80% <= B- < 84% <= B < 87% <= B + <	90%	
70% <= C- < 74% <= C < 77% <= C + <	80%	
60% <= D- < 64% <= D < 67% <= D + <	70%	
	F	< 60%

Exams: Three Midterm Exams are scheduled, on February 17, March 24, and April 28. Each exam will be two hours long, and will take place during the Monday recitation period, 1-3 PM in UTC 4.104. If you need to arrange a different time for an exam, you must notify me at least one week in advance.

Quizzes: Quizzes will be given weekly, at the end of recitation, to assess progress on Simulink coding and on learning the main concepts. Students will be able to drop one quiz from the quiz average.

Homework: Homework will be assigned each week on Monday, and due a week later on Tuesday by midnight. Exceptions may be granted in extraordinary circumstances, but only prior to the deadline. Unless an exemption is granted, late homework turned in within one week of the deadline will only receive 80% credit. You are responsible for verifying that your homework has uploaded properly to Canvas. You are encouraged to take a first pass on the homework problems yourself, and then work with your fellow students to converge on problem solutions. In any case, what you turn in must be your own work.

Knowledge, abilities and skills students should have upon entering this course:

1. A grade of C- or higher in each of CHE 322 (basic understanding of thermodynamics), CHE 354 (ability to establish models for dynamic process operations), CHE 253M (lab experience with sensors, dynamics, and data analysis)
2. Completion of CHE 348 (solve algebraic and ordinary differential equations analytically and numerically and perform basic matrix operations)
3. Ability to write dynamic macroscopic energy and mass balances for chemical systems, and solve them analytically or numerically.
4. Awareness of complex variables and complex functions.
5. Working knowledge of MATLAB and Excel (CHE 210, CHE 317, CHE 348)

Knowledge, abilities, and skills students should gain from this course:

1. Ability to develop first-principles and transfer function based dynamic mathematical models for chemical processes.
2. Ability to analyze process stability and dynamic responses.
3. Ability to empirically determine approximate process dynamics from step response data.
4. Familiarity with different types of PID feedback controllers.
5. Ability to read block diagrams and process and instrumentation diagrams.
6. Ability to design feed forward, ratio, and cascade controllers.
7. Knowledge of multivariable process interactions.
8. Familiarity with Model Predictive Control technology.
9. Apply engineering design principles to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.

Matlab/Simulink Software: You must download MATLAB/Simulink software and install it on your own computer. **Be sure to check the boxes for Simulink and the Control System, System Identification, Optimization, and Symbolic Math toolboxes prior to installation.** Matlab is available free-of-charge to UT students at:

<https://www.mathworks.com/academia/tah-portal/university-of-texas-at-austin-875352.html>

Schedule: Note that this schedule may change slightly as we work our way through the semester. I will provide updates at the beginning of each lecture.

Week #	Week	Chapter(s)	Mo Lecture Topic	Mo Recitation Topic	HW Due Tu 11:59	We Lecture Topic	Fr Lecture Topic
1	13-Jan-2025	1	Intro/Syllabus, 1	matlab/simulink		1	2
2	20-Jan-2025	2/3	MLK-Holiday	1/2	HW1	2	3
3	27-Jan-2025	3/4	3	2/3	HW2	4	4
4	3-Feb-2025	5/6	5	4/5	HW3	5	6
5	10-Feb-2025	6/7	6	5/6	HW4	7	Exam 1 Review
6	17-Feb-2025	8		Exam 1		8	8
7	24-Feb-2025	8/9	8	8	HW5	9	9
8	3-Mar-2025	11/12	11	9/11	HW6	11	12
9	10-Mar-2025	12	12	11/12	HW7	12	Exam 2 Review
10	17-Mar-2025		Spring Break				
11	24-Mar-2025	15		Exam 2		15	15
12	31-Mar-2025	16	16	15/16	HW8	16	16
13	7-Apr-2025	19	19	16/19	HW9	19	19
14	14-Apr-2025	20	20	19/20	HW10	20	20
15	21-Apr-2025	20/10	10	20/10	HW11	10	Exam 3 Review
16	28-Apr-2025			Exam 3			
17	5-May-2025						

Chapter	Chapter Topic	Chapter	Chapter Topic
1	Introduction to process control	9	Control system instrumentation
2	Theoretical models of chemical processes	11	Closed-loop system analysis
3	Laplace transforms	12	PID controller design and tuning
4	Transfer functions	15	Feedforward and ratio control
5	First and second order systems	16	Enhanced single-loop control strategies
6	More complicated processes	19	Real-Time Optimization (RTO)
7	Empirical Models from Process Data	20	Model Predictive Control (MPC)
8	Feedback control (PID algorithm)	10	Process Safety

Communication: Email will be the primary means of communication with members of the instruction team. **Please preface your email subject line with “CHE 360 –” so that we may sort your email appropriately.** Please email me directly, rather than sending a message through Canvas (I cannot easily sort and search email in Canvas).

Professionalism Statement: In the classroom I expect the same level of courtesy, sensitivity, and respect for others that you would find in a professional workplace environment. Courtesy and sensitivity are especially important in a group setting where you may interact with people of different cultures, religions, politics, sexual orientations, genders, gender variances, and nationalities. The goal is for everyone to feel comfortable attending this class, participating in discussions, and asking questions.

Feedback: Your feedback is important to me, and I encourage you to let me know right away if you have strong feelings about anything that we do, positive or negative. My goal is to teach the best possible class, and I believe strongly that there is always room for improvement. You'll have two scheduled opportunities to provide feedback, via the Midterm Course Survey and the final Course Evaluation. Your responses to these will be voluntary, however, I encourage you to provide constructive feedback that will allow me to improve the class.

Recitations: The Teaching Assistant will conduct recitations, where MATLAB/Simulink syntax and coding will be reviewed and problems like those on the HWs will be discussed and solved. This is an excellent opportunity to strengthen your programming skills and get ahead on the HWs.

Regrades: All regrade requests for exams and homework must be submitted by email within one week of receiving your graded exam or homework. Homework regrade requests should be directed to the grader, with the TA CC'd. Exam regrade requests should be directed to the Instructor.

Electronics: At the start of class, please set your cell phone to silent and put it away. You are encouraged to bring your laptop to lectures, as we will be writing code together on most class days. However, its primary use should be to follow along with programming exercises and not to be a distraction to yourself and to those around you.

University of Texas Honor Code: The core values of The University of Texas at Austin are learning, discovery, freedom, leadership, individual opportunity, and responsibility. Each member of the university is expected to uphold these values through integrity, honesty, trust, fairness, and respect toward peers and community.

Cheating: As with plagiarism, students who are caught cheating will receive no credit for the assignment or exam that they cheated on, and the incident will be reported to the office of Student Conduct and Academic Integrity without hesitation or prior warning. The referral will be accompanied by a recommended sanction that may be as severe as a one-letter grade reduction in the final course grade. Once accused of academic misconduct in a course you cannot drop that course. Please be proactive about avoiding the appearance of cheating. **Please note that exams are scanned when they are turned in, and also after they have been graded.**

Disability Statement: The university is committed to creating an accessible and inclusive learning environment consistent with university policy and federal and state law. Please let me know if you experience any barriers to learning so I can work with you to ensure you have equal opportunity to participate fully in this course. If you are a student with a disability, or think you may have a disability, please contact Services for Students with Disabilities (SSD). Please refer to SSD's website for contact and more information: <http://diversity.utexas.edu/disability/>. If you are already registered with SSD, please deliver your Accommodation Letter to me **within the first week** of the semester so we can discuss your approved accommodations and needs in this course.

Behavior Concerns Advice Line (BCAL): If you are worried about someone who is acting differently, you may use the Behavior Concerns Advice Line to discuss by phone your concerns about another individual's behavior. This service is provided through a partnership among the Office of the Dean of Students, the Counseling and Mental Health Center (CMHC), the Employee

Assistance Program (EAP), and The University of Texas Police Department (UTPD). Call 512-232-5050 or visit <http://www.utexas.edu/safety/bcal>.

Sharing of Course Materials is Prohibited: No materials used in this class, including, but not limited to, lecture hand-outs, videos, assessments (quizzes, exams, papers, projects, homework assignments), in-class materials, review sheets, and additional problem sets, may be shared online or with anyone outside of the class unless you have my explicit, written permission. Unauthorized sharing of materials promotes cheating, and is a violation of the University's Student Honor Code and an act of academic dishonesty. I am well aware of the sites used for sharing materials, and any materials found online that are associated with you, or any suspected unauthorized sharing of materials, will be reported to the office of Student Conduct and Academic Integrity.

Class Recordings: Class recordings are reserved only for students in this class for educational purposes and are protected under the Family Educational Rights and Privacy Act (FERPA). The recordings should not be shared outside the class in any form. Violation of this restriction by a student could lead to Student Misconduct proceedings.